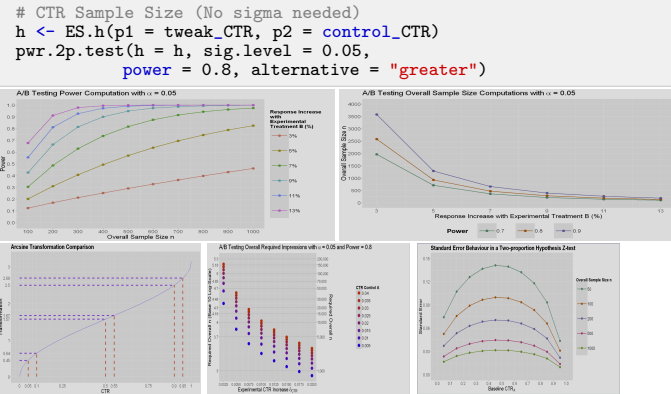


1 Continuous Response Analysis
1.1 Power and Sample Size Dynamics

- Sample size computation for continuous outcomes utilizes pwr.t.test(). The required overall sample size n is driven by three main factors: significance (alpha), power (1 - beta), and effect difference (delta).
- Increasing Power: To reduce the probability of a Type II error, n must increase.
 - Decreasing Delta (delta): If the anticipated percentage increase is small (e.g., 3% vs 13%), n must increase dramatically because the signal is harder to distinguish from noise.
 - Plotting Significance: Steeper curves in power plots indicate that required sample size increases rapidly as higher power levels are demanded for small delta.
 - Statistical: Detecting smaller effects requires larger n.
 - Practical/Business: Balancing the cost of exposing users to a potentially worse version vs. the cost of failing to detect a real improvement.

2 Click-Through Rate (CTR) Analysis
2.1 Proportion-Based Power

- CTR is a binary response measured as a proportion. Unlike continuous data, CTR analysis does not require computing sigma or a treatment mean because variation is inherently tied to the proportion itself.
- Arcsine Transformation: The function ES.h() transforms proportions into an effect size h [10]. This transformation accounts for the fact that standard error (SE) varies across the range.
 - Baseline CTR_A Impact: SE approaches a maximum near 0.5. Consequently, if CTRs are jointly around 0.5, n must increase to overcome high uncertainty.
 - Extreme Baselines: If CTR baselines are near 0 or 1, SE is smaller, allowing for a smaller required n.



3 Early Stopping and Peeking
3.1 Aggressive Peeking

- Peeking involves accessing data in real-time before reaching n_max.
- Type I Error Inflation: Aggressive peeking (checking every visitor) severely inflates the Type I error rate (up to ~25% for alpha = 0.05) because it gives the test multiple chances to cross the significance threshold by chance.
 - Power Loss: While it can stop early for large effects, aggressive peeking causes a power decrease if the true CTR increase is small.
- 3.2 Principled Peeking
- Principled peeking uses a fixed number of peeks (m) and the Bonferroni correction (alpha/m).
- Error Control: The alpha/m adjustment prevents the inflation of false positives seen in aggressive peeking.
 - Optimal Strategy: 2 principled peeks are often considered ideal; increasing peeks beyond this can significantly reduce power.

Decision	H ₀ is True	H ₀ is False
Fail to Reject	Correct (TN)	Type II Error (FN / beta)
Reject H ₀	Type I Error (FP / alpha)	Correct (TP / Power)

4 Observational Studies & Confounding

4.1 Causal Inference Challenges

In observational studies, the researcher lacks control over the variable of interest X, as subjects often self-select into treatment groups. Because randomization is absent, these studies cannot automatically eliminate the effects of confounding variables, making causal interpretation difficult. The general strategy involves recording potential confounders during data collection and incorporating them into the analysis while tempering causal claims. Data science applications, such as pharmaco-epidemiology, utilize massive healthcare databases to estimate what would occur in a randomized trial setting.

Naive vs. Adjusted: The "direction" of an effect can flip in an adjusted model. This happens because the naive model mixes the relationship between X, Y, and the confounder. Adjusting allows the model to interpret the effect conditional on the confounder.

5 Statistical Metrics for Binary Outcomes
5.1 Log-Odds and Odds Ratios

Binary logistic regression utilizes the log-odds, defined as the natural logarithm of the ratio of success probability to failure probability. The Odds Ratio (OR) measures the association or potential causation between an exposure X and an outcome Y. An OR = 1 indicates the exposure does not affect the outcome odds, while an OR > 1 suggests the exposure is associated with higher odds of the outcome. The Western Collaborative Group Study (WCGS) uses these metrics to assess if Type A behavior patterns lead to higher odds of coronary heart disease (CHD).

```
initial_bin_log <- glm(chd69 ~ dibpat, data = training_data, family = "binomial")
tidy(initial_bin_log, conf.int = 0.95)
detach(training_data)
training_data <- training_data %>%
  mutate(bmi_bins = cut(bmi, breaks = c(min(bmi), quantile(bmi, (1:3) / 4), max(bmi))), include.lowest = TRUE))
levels(training_data$bmi_bins)
bin_log_model <- glm(chd69 ~ dibpat + age_bins + bmi_bins + smoke, data = training_data, family = "binomial")
tidy(bin_log_model, conf.int = 0.95) %>%
  mutate(exp.estimate = exp(estimate)) %>%
  mutate_if(is.numeric, round, 3)
p_values <- c(anova(bin_log_model, triple_bin_log_model, test = "LRT"))$`Pr(>Chi)`[2]) # Storing p-value
```

6 Stratification and Modeling Strategies
6.1 Adjusting for Multiple Confounders

Stratification involves "slicing" data based on confounding variables, such as age bins, to identify stratum-specific effects. However, stratifying by multiple variables (e.g., age, smoking, and BMI) can lead to very small subsamples, resulting in insufficient data for estimation and high uncertainty. An overall Binomial Logistic Regression serves as a more efficient approach, acting as a "proxy for blocking" by including confounders as standalone regressors. This model assumes a simple structure for log-OR variation across strata and the absence of interactions between the exposure and confounders.

7 Model Selection and Assumptions
7.1 Testing and Causal Reasoning

Causal inference from regression models requires meeting strong assumptions, including the inclusion of all relevant confounders and the lack of unobserved confounding. Model selection often employs Likelihood Ratio Tests (LRT) to compare nested models, such as testing for the necessity of interaction terms. While multiple testing corrections like Bonferroni manage significance levels, they can be overly strict and may exclude genuine confounders that are causally relevant. Therefore, model selection should combine statistical testing with subject-matter knowledge about the data-generating process.

8 Sampling Schemes & Proxy Ground Truth
8.1 Simulation Frameworks

In frequentist statistics, access to the population's ground truth is often impossible. Simulation serves as a proxy ground truth by using a representative dataset to fit a model, which then generates artificial outcomes (Y) based on estimated regression coefficients and design matrices. This approach allows researchers to assess the accuracy and precision of different sampling designs before implementation. Causal inference in these models relies on controlling for confounders (C_j) to assume a common effect of X over Y across all subjects.

Proxy vs. Synthetic: Proxy ground truth (built from real surveys) preserves realistic associations between X, Y, and confounders, whereas simple synthetic data often misses these complexities and

yields overly idealized performance estimates.

9 Comparing Observational Schemes
9.1 CS, CC, and CO Temporalities

Three primary schemes exist: 1. Cross-Sectional (CS) is contemporaneous, taking a simple random snapshot of the population. 2. Case-Control (CC) is retrospective, sampling half from those with the outcome (Y = 1, cases) and half from those without (Y = 0, controls). 3. Cohort (CO) is prospective, sampling based on exposure (X) and tracking outcomes over time. While CS is ideal for early research, CC is significantly more efficient when the outcome is rare, offering higher power and lower RMSE.

```
# Simulating outcomes for Proxy Ground Truth
pred_prob <- function(logit) {exp(logit) / (1 + exp(logit))}
sim_pop$chd69 <- rbinom(n = nrow(sim_pop), size = 1,
  prob = pred_prob(logit = model_matrix %*% coefs))
# Case-Control Sampling (Retrospective)
CC_subjects <- c(
  sample(which(sim_pop$chd69 == "0"), size = n / 2),
  sample(which(sim_pop$chd69 == "1"), size = n / 2))
```

10 Case-Control Efficiency
10.1 Case:Control Ratios and Precision

The efficiency of Case-Control studies is analyzed via a modified power analysis focusing on the Standard Error (SE) of the log-OR estimate. In populations where the event is rare (e.g., < 10%), oversampling cases (Y = 1) and undersampling controls (Y = 0) acts as a winning strategy for precision. Precision increases as the Case:Control ratio reaches or slightly exceeds 1.0, particularly when the estimated log-OR is low. The SE is minimized as the cell frequencies in the contingency table become more balanced through this strategic oversampling.

11 Matched Case-Control Studies
11.1 Matching and McNemar Analysis

Matched Case-Control (CC) sampling involves selecting n/2 cases (Y = 1) and then finding n/2 controls (Y = 0) that exactly match each case's confounding strata (C_j). This design allows for stronger causal claims. Because pairs are dependent, Binary Logistic Regression via glm() can lead to the "sparse data problem". Instead, the McNemar test is used, focusing on discordant pairs: n_1,0 (Case X = 1, Control X = 0) and n_0,1 (Case X = 0, Control X = 1). The OR is estimated as n_1,0/n_0,1. Simulation shows matched CC has smaller BIAS and RMSE than unmatched CC at small sample sizes (e.g., n = 100).

Why not glm?: Classical glm() assumes independent observations. Matched data has a pairing structure (dependency). Using glm() on matched data leads to the "sparse data problem".

```
# Estimating log-OR for matched samples
logOR_est <- log(n10) - log(n01)
logOR_se <- sqrt(1/n10 + 1/n01)

# Ordinal Contrasts
options(contrasts = c("contr.treatment", "contr.sdif"))
```

12 Ordinal Regressors
12.1 Polynomial and Successive Contrasts

- Ordinal regressors are categorical factors with ordered levels (e.g., discount percentages). Rather than standard dummy coding, contrasts (vectors summing to zero) provide deeper insight.
- Polynomial Contrasts: Decompose data into Linear (L), Quadratic (Q), and Cubic (C) trends. Useful for testing if a specific trend exists in the ordered levels.
 - Successive Differences (contr.sdif): Directly compare the mean of one level to its immediate predecessor (e.g., L_2 - L_1). These are often more interpretable for specific statistical inquiries.

Decision Strategy:

- CS: Quick preliminary evidence.
- CC: Efficient for rare outcomes (start with diseased).
- CO: Efficient for rare exposures (start with exposed).

12.2 Experimental vs. Observational

Experimental: Full control over treatments. Randomization eliminates confounders. Blocking ensures precision by grouping homogeneous units before randomizing.

Observational: No randomization. Uses stratification or matching via confounders to approximate causal comparisons. Analysis must reflect collection: CS (simple random), CC (retrospective, rare outcomes), CO (prospective).